

Chronotype moderates the association between adverse childhood events and adult insomnia severity

Grant Benham

The University of Texas
Rio Grande Valley

Methods

Five hundred and seventy-one undergraduates (76% female; 95% Hispanic; mean age = 21.1) completed an online anonymous survey that included self-report measures of ACEs (Adverse Childhood Experiences Questionnaire) and past-month insomnia severity (Insomnia Severity Index), along with a single item assessing chronotype. A moderation analysis was conducted using Hayes' PROCESS Macro V4.1 with 5,000 bootstrapped samples. ACEs were examined as a predictor of insomnia severity, with extreme chronotypes (definitely morning/evening) as a moderator, controlling for both age and gender..

Discussion

These findings support the association between ACEs and insomnia severity in a young adult population and is the first study to demonstrate that the observed association is moderated by chronotype. Within our sample, the association was observed in those who reported definite evening chronotypes, but not in those reporting definite morning chronotypes. Clinicians should therefore consider the role that evening chronotype might have in amplifying the potential role of ACEs as a risk factor for insomnia. Our study was limited to retrospective reports of ACEs and the nature of our sample limits generalizability. However, the issue of insomnia is an important one to address in these young adults, given that the prevalence of insomnia may be greater in college students than in the general population. Given that insomnia is proposed as a risk factor for psychological and medical disorders, not merely a consequence or comorbidity, this study aids in identifying potential at risk populations.

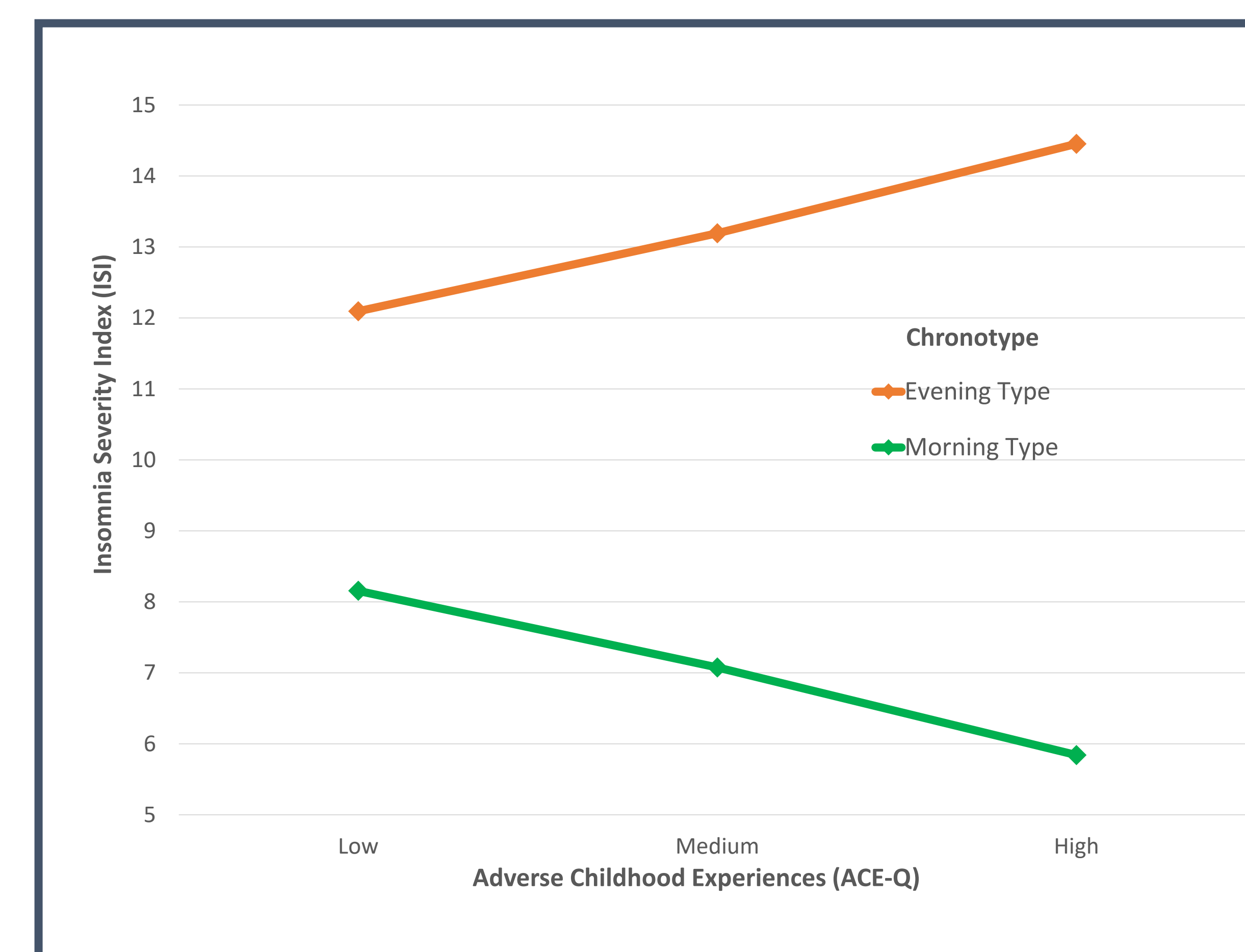
Introduction

Sleep difficulties are common in college students and linked to poorer mental and physical health. Psychological stress is recognized as a strong predictor of sleep difficulties, with the large majority of studies focusing on recent stressors or perceived stress. More recently, researchers have found an association between early life stressors (adverse childhood experiences [ACEs]) and sleep difficulties in adulthood.

Research on the association between ACEs and insomnia is limited and inconsistent. Furthermore, little is known about factors that might moderate this association. We test the ACEs-insomnia association and examine whether chronotype (evening type vs morning type) moderates this association.

Results

Two-thirds of the sample (67.3%) reported having experienced one or more ACEs and 20.7% reported having experienced four or more. 57% met the threshold for insomnia, based on an insomnia severity index score of 10 or greater. **ACEs were significantly correlated with insomnia severity** $r(569) = .24$, $p < .001$. This relationship remained statistically significant when controlling for age and sex (partial correlation $r(567) = .22$, $p < .001$. Chronotype significantly moderated this association (interaction effect: $B = 1.19$, $p = .046$, 95% CI [.02, 2.36]). **Conditional effects for each chronotype indicated that the association was only statistically significant for definite evening types** (conditional effect $B = .60$, $p = .010$, 95% CI [.15, 1.05]), not for morning types (conditional effect $B = -.59$, $p = .29$, 95% CI [-1.68, .50]).



Contact

Dr. Grant Benham
grant.benham@utrgv.edu

